

Secure Remote Access to Synology Photos via Tailscale

A complete step-by-step guide to setting up Tailscale on your Synology NAS so you can securely access and sync Synology Photos from your iPhone — no port forwarding, no public IP, no Synology relay servers touching your data.

What You'll Need

- A Synology NAS running **DSM 7.0 or later**
 - **Synology Photos** installed on your NAS
 - An **iPhone** with the Synology Photos iOS app
 - A **Tailscale account** (free for personal use — up to 100 devices)
 - **Admin access** to your NAS
 - A computer or phone on the **same local network** as the NAS (for initial setup)
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Part 1: Install Tailscale on Your Synology NAS

Step 1 — Open the Package Center

1. Log in to your DSM web interface (e.g., `http://192.168.1.100:5000`).
2. Open **Package Center** from the main menu.

Step 2 — Install Tailscale

1. In the Package Center search bar, type **Tailscale**.
2. Click **Install**. If Tailscale doesn't appear in search results, you can install it manually:
 - Go to <https://pkgs.tailscale.com/stable/#spks>
 - Download the `.spk` file matching your NAS architecture (most modern "+" models like DS224+, DS923+, DS423+ use **x86_64**; check **Control Panel** → **Info Center** if unsure).
 - In Package Center, click **Manual Install** (top-right), browse to the downloaded `.spk`, and follow the prompts.
3. Accept any warnings about third-party packages.

Step 3 — Authenticate Tailscale

1. Once installed, click **Open** on the Tailscale package.
2. A new browser tab opens with a **Log In** button.
3. Click it and sign in with your identity provider (Google, Microsoft, GitHub, Apple, etc.).
4. Authorize the NAS to join your Tailscale network (called a "tailnet").
5. After authentication, your NAS will appear in your Tailscale admin console at <https://login.tailscale.com/admin/machines> with an assigned IP in the `100.x.x.x` range.

Write down your NAS's Tailscale IP (e.g., `100.64.0.1`). You'll need it later.

Part 2: Enable Outbound Connections (Required for DSM 7)

DSM 7 restricts what packages can do by default. Tailscale can receive inbound connections, but other apps on your NAS (like Synology Photos) won't be able to reach Tailscale devices unless you run a startup script. This step is important for full functionality.

Step 4 — Create a Boot-Up Task

1. Go to **Control Panel** → **Task Scheduler**.
2. Click **Create** → **Triggered Task** → **User-defined script**.
3. In the **General** tab:
 - **Task name:** `Tailscale TUN Setup` (or any name you like)
 - **User:** `root`
 - **Event:** `Boot-up`
 - Make sure **Enabled** is checked.
4. Click the **Task Settings** tab.
5. In the **User-defined script** box, paste this command:

```
/var/packages/Tailscale/target/bin/tailscale configure-host; synosystemctl restart pkgctl-Tailscale.service
```

6. Click **OK**. You'll see a warning about running scripts as root — confirm it.
7. **Reboot your NAS** (or right-click the task and select **Run** to execute it immediately).

Why this matters: This script enables TUN device creation, allowing outbound Tailscale connections from DSM services. Without it, some features may not work properly.

Part 3: (Optional but Recommended) Advertise Your Local Subnet

This step lets you access your NAS (and other devices on your home network) using their regular local IP addresses through Tailscale, rather than only the Tailscale `100.x.x.x` address. This makes things more seamless.

Step 5 — Enable Subnet Routing

Option A — Via Task Scheduler (no SSH needed):

1. Go to **Control Panel** → **Task Scheduler**.
2. Click **Create** → **Scheduled Task** → **User-defined script**.
3. In the **General** tab:
 - **Task name:** `Tailscale Subnet Routes`
 - **User:** `root`
4. In the **Schedule** tab, set it to **Do not repeat** (you'll run it manually once).
5. In **Task Settings**, paste:

```
/var/packages/Tailscale/target/bin/tailscale up --advertise-routes=192.168.1.0/24  
--advertise-exit-node --reset
```

Important: Replace `192.168.1.0/24` with your actual local subnet. If your NAS IP is `192.168.0.100`, use `192.168.0.0/24`. If it's `10.0.0.50`, use `10.0.0.0/24`.

6. Click **OK**, then right-click the task and select **Run**.

Option B — Via SSH:

1. Enable SSH: **Control Panel** → **Terminal & SNMP** → **Enable SSH service**.
2. SSH into your NAS: `ssh admin@192.168.1.100` (use your admin username).
3. Run:

```
bash  
  
sudo tailscale up --advertise-routes=192.168.1.0/24 --advertise-exit-node --reset
```

Step 6 — Approve Routes in the Tailscale Admin Console

1. Go to <https://login.tailscale.com/admin/machines>.
 2. Find your NAS in the machine list.
 3. Click the **three dots (...)** next to it → **Edit route settings**.
 4. **Enable** the subnet route (e.g., `192.168.1.0/24`).
 5. Optionally enable **Use as exit node** if you want to route all your phone's internet traffic through your home connection when away.
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Part 4: Install Tailscale on Your iPhone

Step 7 — Set Up Tailscale on iOS

1. Download **Tailscale** from the App Store.
2. Open it and tap **Sign in**.
3. Log in with the **same identity provider** you used for your NAS (Google, Microsoft, etc.).
4. iOS will ask to allow a VPN configuration — tap **Allow**.
5. Once connected, you'll see a toggle to enable/disable the Tailscale VPN. Turn it **on**.
6. Your iPhone is now part of the same tailnet as your NAS.

Tip: In Tailscale's iOS settings, you can enable **Always On VPN** so you don't have to manually toggle it each time.

Part 5: Configure Synology Photos to Use Tailscale

Step 8 — Make Sure Synology Photos Is Set Up on Your NAS

1. In DSM, open **Package Center** and confirm **Synology Photos** is installed and running.
2. Open Synology Photos in DSM and make sure your photo libraries are set up (Personal Space enabled, Shared Space configured if needed).
3. If you haven't already, enable photo backup settings for the users who will connect.

Step 9 — Connect the Synology Photos iOS App via Tailscale

1. On your iPhone, make sure **Tailscale is connected** (green toggle in the Tailscale app).
2. Open the **Synology Photos** app on your iPhone.
3. When it asks you to log in or add a server, enter your **NAS's Tailscale IP address**:

`https://100.x.x.x:5001`

Replace `100.x.x.x` with the actual Tailscale IP you noted in Step 3. Port `5001` is the default HTTPS port for DSM.

Alternative: If you set up subnet routing in Part 3, you can also use your NAS's regular local IP (e.g., `https://192.168.1.100:5001`), which will work both at home and remotely through Tailscale.

4. Log in with your Synology DSM credentials.
5. You should now see your photo library.

Step 10 — Enable Photo Backup

1. In the Synology Photos iOS app, go to **Settings** (tap the profile/gear icon).
2. Tap **Photo Backup** and enable it.
3. Configure your preferences:
 - **Backup rule:** "Back up all photos" (for first-time setup) or "Back up new photos only"
 - **Wi-Fi only:** Your choice — Tailscale traffic is encrypted regardless of network
 - **Backup destination:** Choose your preferred folder (Personal Space or a custom folder)
4. Tap **Done**. The app will begin uploading photos to your NAS.

iOS background limitation: iOS may pause the backup when the app goes to the background. To speed up large initial uploads, keep the app open in the foreground, or use the **Focused Backup** feature (available in recent app versions) which keeps the upload active.

Part 6: Disable QuickConnect (Optional but Recommended)

Now that Tailscale handles remote access, you can disable QuickConnect to minimize your NAS's exposure.

Step 11 — Turn Off QuickConnect

1. Go to **Control Panel** → **QuickConnect** (or **External Access** → **QuickConnect**).
2. Uncheck **Enable QuickConnect**.
3. Click **Apply**.

Your NAS is now only accessible from your local network or through Tailscale — not discoverable through Synology's servers.

Part 7: Firewall Hardening (Optional)

If you have DSM's built-in firewall enabled, you need to allow Tailscale traffic through it.

Step 12 — Add a Firewall Exception

1. Go to **Control Panel** → **Security** → **Firewall**.
 2. Click **Edit Rules** on your active profile.
 3. Add a rule:
 - **Source IP:** Specific IP → Subnet: `100.64.0.0` / Mask: `255.192.0.0`
 - **Ports:** All
 - **Action:** Allow
 4. Make sure this rule is **above** any deny-all rule.
 5. Click **Apply**.
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Verification & Troubleshooting

Test Your Setup

1. **Disconnect from your home Wi-Fi** (use mobile data).
2. Open the **Tailscale** app on your iPhone and make sure it's connected.
3. Open the **Synology Photos** app — you should be able to browse your library and upload new photos.
4. To confirm you're going through Tailscale: try accessing `https://100.x.x.x:5001` in Safari — you should see the DSM login page.

Common Issues

Problem	Solution
Synology Photos can't connect remotely	Make sure Tailscale is toggled on on your iPhone. Check that you're using the correct Tailscale IP and port (<code>100.x.x.x:5001</code>).
"Connection refused" errors	Verify the boot-up task from Step 4 has run. Try rebooting the NAS.
Slow connection	Check the Tailscale admin console — if it says "relay" instead of "direct", your NAT traversal isn't working optimally. This is rare but can happen behind certain strict firewalls.
NAS not showing in Tailscale admin	Re-open the Tailscale package in DSM, click Log In again, and re-authenticate.
Can't reach local IP through Tailscale	Make sure you approved the subnet route in the admin console (Step 6).
Backup pauses in background (iOS)	This is an iOS limitation. Open the app periodically, or use Focused Backup to keep it active during large transfers.

Quick Reference

Item	Value
Tailscale admin console	https://login.tailscale.com/admin
NAS Tailscale IP	<code>100.x.x.x</code> (find in admin console)
DSM web interface via Tailscale	<code>https://100.x.x.x:5001</code>
Synology Photos via Tailscale	Use the Tailscale IP in the iOS app login
Boot-up script	<code>/var/packages/Tailscale/target/bin/tailscale configure-host; synosystemctl restart pkgctl-Tailscale.service</code>

Security Summary

With this setup:

- **No ports** are open on your router
- **No public IP** is required
- **No Synology relay servers** touch your data
- All traffic is **end-to-end encrypted** with WireGuard (keys held only by your devices)
- Access requires authentication via your identity provider + Tailscale + DSM credentials — **three layers**
- Even Tailscale's coordination servers **cannot decrypt** your traffic

This gives you a self-hosted, private photo cloud with security that exceeds what QuickConnect offers — while remaining easy to use day-to-day.